

INTRO TO R

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About me



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Background

- PhD (Public Health Epidemiology) from USM, 2024
- MSc (Medical Statistics) from USM, 2019
- MBBCh from Al-Azhar University, 2015

Interest:

- Medical statistics, meta-analysis, bibliometrics
- Machine learning and deep learning application in medical sciences
- Application of R on health/medical data

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Material

https://github.com/tengku-hanis/intro2r_postgrad_unisza.git

Learning outcome

- Not to know everything about machine learning and R
- To know enough so that you can further explore and learn on your own

Tentative

Time	Agenda
09.00 - 09.15 am	Registration
09.15 - 09.45 am	Intro to R, RStudio, and Posit Cloud
09.45 - 10.45 am	R basics
10.45 - 12.00 pm	Data wrangling
12.00 - 1.00 pm	Data visualisation
01.00 - 02.00 pm	Solat, rest, and lunch
02.00 - 03.00 pm	Reproducible data analysis using Quarto
03.00 - 04.00 pm	The use of AI in R
04.00 pm	End

Content

- Why use R
- R, RStudio, and Posit
- Installation
- Flow of data analysis
- Getting help
- Common beginner errors
- Tips
- Suggested reading
- Links

Why use R

- Flexibility to use R everywhere - R is free
- Wide functionalities:
  - Statistical analysis
  - Machine learning, deep learning
  - Image processing (to certain extent)
  - Website development
  - Etc
- Wide range of packages (for example, in CRAN - 25,709 official packages):
  - Wide range of statistical analysis: basic to advanced
  - Early access to new statistical method
  - All packages are free



R, RStudio, and Posit

- R:
  - Programming and statistical software
  - Run in the back of IDE
- RStudio:
  - Integrated development environment (IDE)
  - Run on front, which the user interact with
- Posit:
  - Previously, also known as RStudio
  - The company that build RStudio





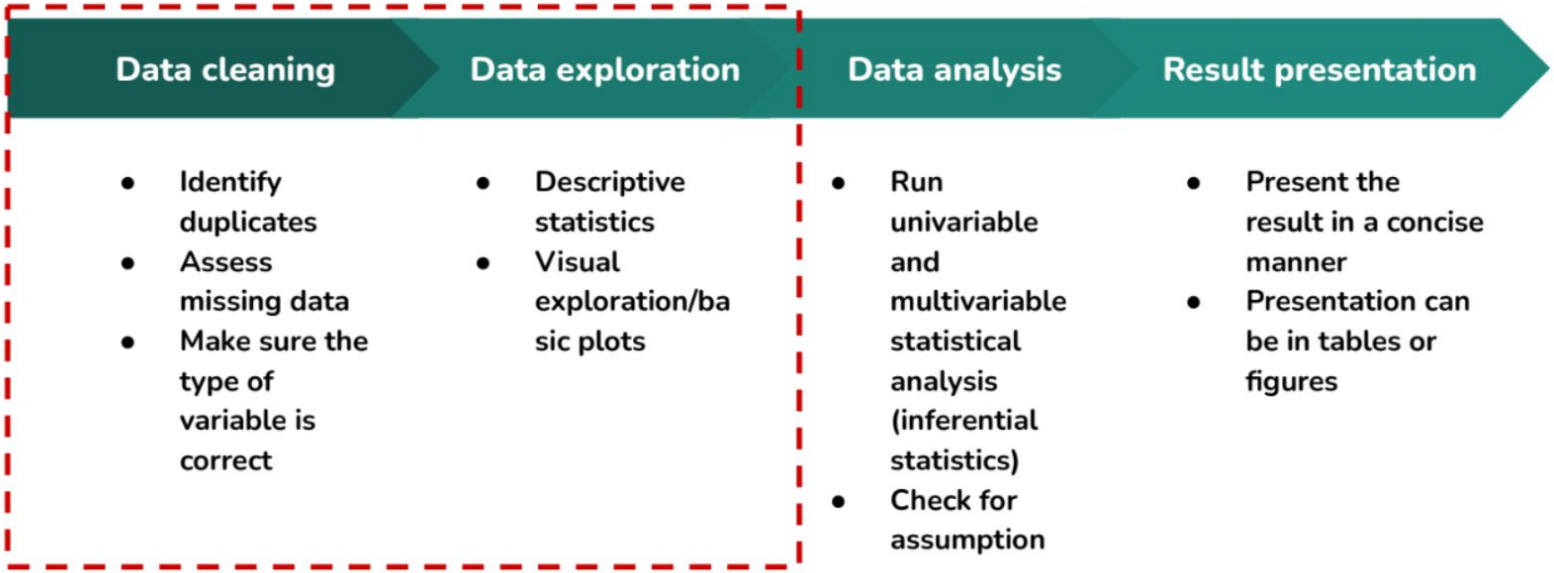


- Other IDEs for R:
  - Google Collab (cloud-based)
  - Posit Cloud (cloud-based)
  - PyCharm
  - VS code - Visual Studio Code
  - Positron
  - And many more
- Why use RStudio:
  - Supported by Posit
  - Probably one of the most (if not the most) popular IDE for R
  - Large support community

Installation

- Posit Cloud (we will be using this throughout this workshop):
  - Cloud version of RStudio
  - No need installation
  - Free up to an extent
  - Need to make an account
- RStudio:
  - Install base R first
  - Install RStudio

Flow of data analysis



Getting help

- ?mean or help(mean)
- Scroll to Examples and run them
- Stack Overflow, community forums
- AI assistants (ChatGPT, Copilot): useful, but verify the answer

Common beginner errors

- R is case-sensitive: mean is not Mean
- Missing bracket or quote
- Could not find function: package not loaded (use library())
- Read the error message; it usually tells you the line

Tips

- It takes some time before you going to be comfortable and familiar with R
- The more you use R, the faster you become familiar with R





Suggested readings/references

- Hanis, T. M. (2026, January 15). R essentials: A beginner's Guide to Data Analysis. https://tengku-hanis.github.io/r_essential/

Links

- Install R and RStudio on Mac OS - https://www.youtube.com/watch?v=1PsPfMaLWSk&t=46s
- Install R and RStudio on Windows - https://www.youtube.com/watch?v=h2IPWVXaUuU



Any question?